

SPRAY DYNAMICS

SCARF FEEDER®

GENERAL DESCRIPTION

Introduction

The Scarf Feeder is designed to distribute an even flow of dry seasoning to the coating zone with simple controls and minimal user interaction during change over. The spreading action is essential to uniform coating and is achieved from the bias cut tray of this vibratory conveyor.

Construction

This equipment is constructed of 304 Stainless Steel, using sanitary design criteria. All components are designed and mounted for ease of maintenance and maximum flexibility.

The Scarf Feeder assembly consists of the pan, base, fiberglass leaf springs, drive coil (motor), armature, accelerometer, frequency amplitude controller, isolation mounts, and an optional cover.

The drive coil is electromagnetic and is completely sealed in epoxy; this drive coil is mounted to the base. The pan is designed to convey and distribute product; this pan is mounted to the base with leaf springs. The assembly is designed to run at a frequency near the natural frequency. The armature is mounted on the pan; the distance between the armature and the drive coil is adjustable to run the conveyor with minimum current and without striking. The drive coil pulls and releases the armature attached to the pan causing the pan to vibrate, constrained by the leaf springs, to run at 20 degrees to the vertical. This allows material in the pan to transfer from infeed to discharge in a series of "hops".

The controller allows variation of driving frequency and amplitude of vibration to operate the conveyor near resonance. Drive frequency and air gap (between armature and drive) are set at the factory for optimum performance. As the conveyor operates and components wear and hardware loosen the natural frequency of the system may change. It is therefore important to have a variable frequency controller to tune the system without having to change leaf springs. The accelerometer allows the controller to properly tune the frequency when the Auto Frequency Search is activated within the controller.

In normal use the only adjustment an operator may need to make is the amplitude of vibration.

NOTE:



UNDER NO CIRCUMSTANCES SHOULD THE PAN BE ALLOWED TO VIBRATE MORE THAN 0.1" AMPLITUDE AND THE DRIVE CURRENT SHALL NOT BE ALLOWED TO EXCEED 1 AMP.

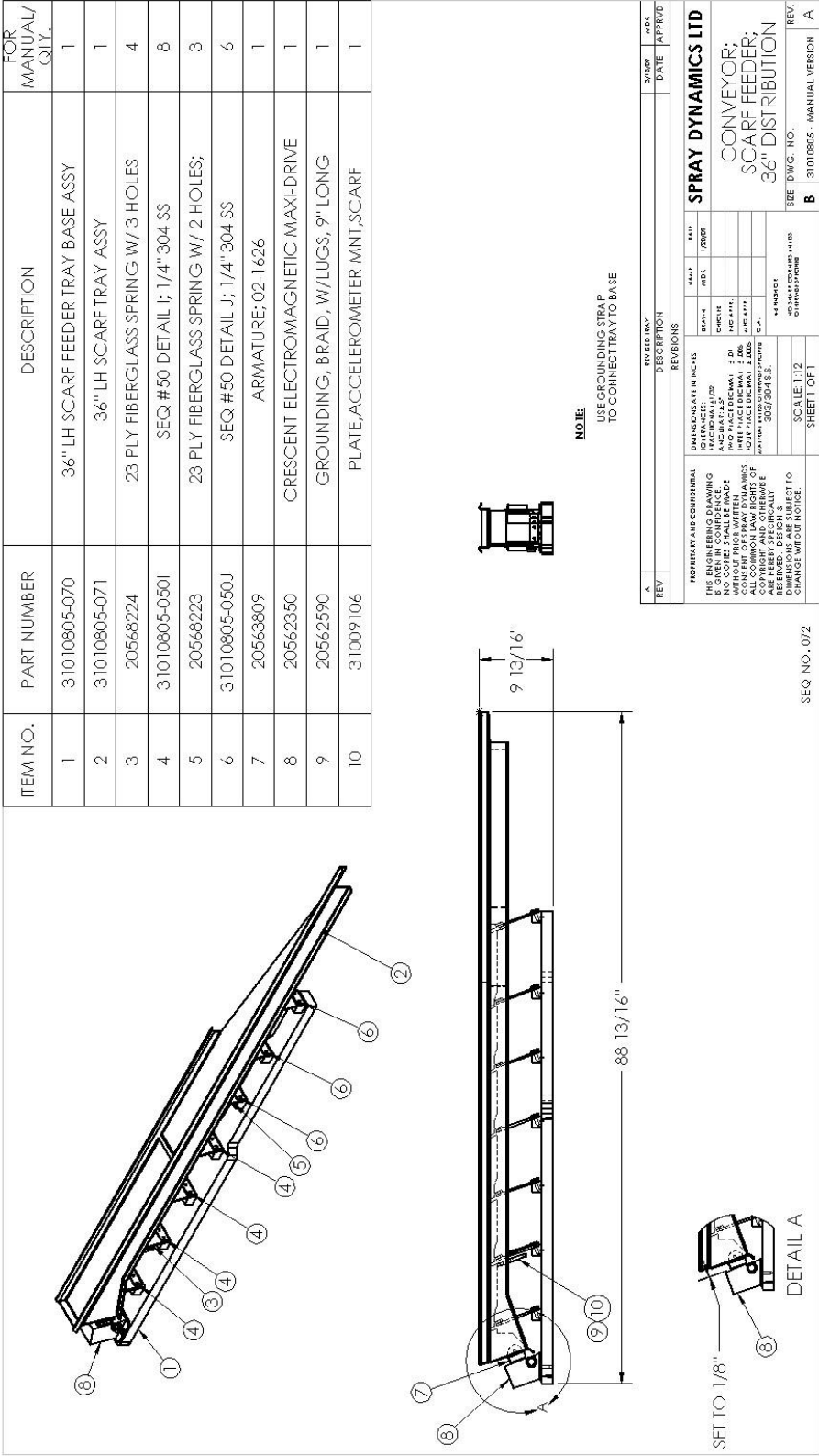
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This equipment is not intended to apply any type of flammable, hazardous, or any material outside the scope of the original contract.

Any variations in the design of this system will void any implied warranty, unless authorized by Spray Dynamics/ A Heat and Control Company. Any new products to be coated must be assessed for the potential risks and hazards and suitable measures must be taken.

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The following paragraphs describe the system in terms of the engineering drawings furnished with this manual. Refer to those drawings as required.

System Framework

This assembly supports other elements of the system. The assembly, made of 304 stainless steel, is mounted on locking casters or bolt pads. It may also include mounting for a Unispense Dry Ingredient Feeder that meters seasoning onto the scarf feeder.

Scarf Feeder

Spray Dynamics Scarf Feeders are made in a variety of sizes; but all conform to a common design consisting of an electromagnetic drive, a steel laminated armature, a leaf spring system, a base frame, a pan to move and distribute product, an accelerometer, and a set of anti-vibration mounts.

Controls

The Controls may be in a panel mounted to the framework or they will be integrated into a consolidated control panel for several pieces of Spray Dynamics equipment.



If work is required inside the control panel, proper Lock-Out/Tag-Out procedures should be followed.

Circuits and controls include:

- System **START/STOP**